Questions

Q1.

A student investigates how the angle of refraction varies with the angle of incidence when light enters a glass block.

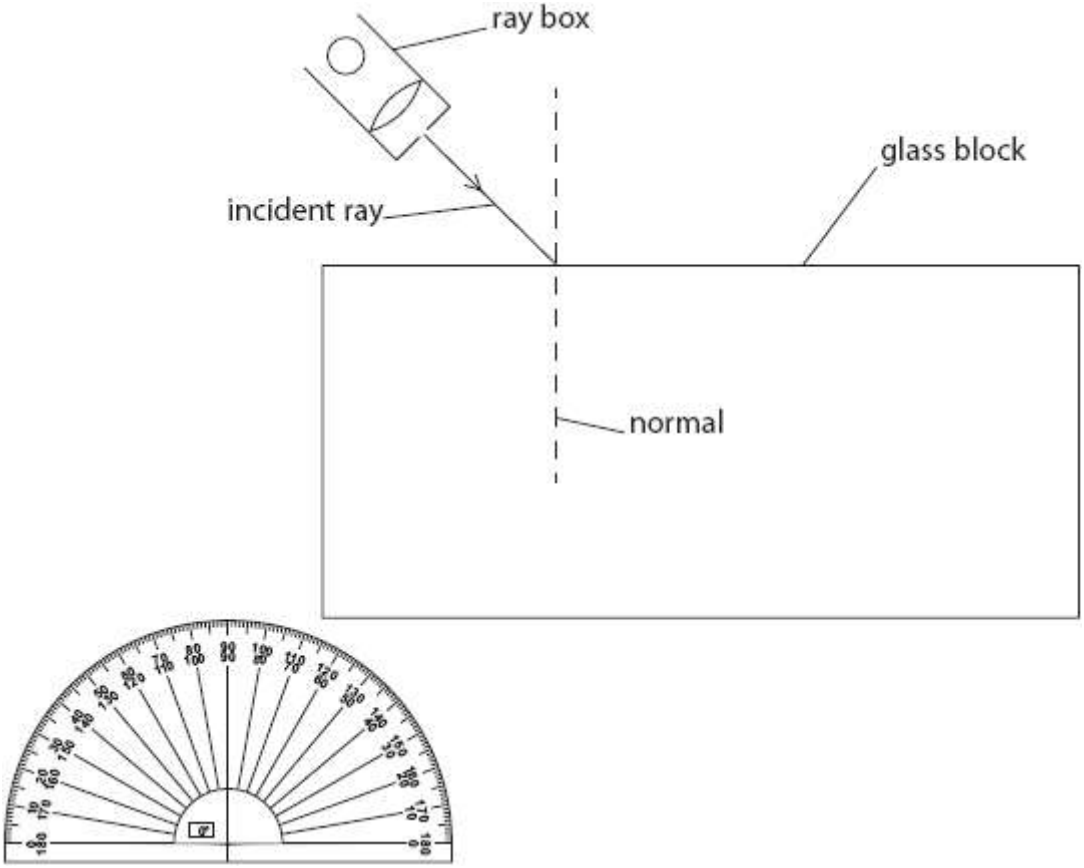

Figure 5

Describe how the student should perform this investigation.

(4)

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(Total for question = 4 marks)

Q2.

Figure 7 shows some apparatus.

Figure 7

Describe an investigation to find out how the nature of a surface affects the amount of thermal energy absorbed by the surface.

You should use the apparatus in Figure 7 and any additional items you choose.
Each can in Figure 7 has a bung in the top with a hole in it.

You may use a diagram if it helps your answer.

(6)

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(Total for question = 6 marks)

Q3.

A student is in a laboratory that has windows.

The student is given a converging lens, and a sheet of paper.

Describe how the student can produce an image of the window frame on the sheet of paper.

(2)

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(Total for question = 2 marks)

Q4.

Figure 1 shows how the brightness of a source of light changes with wavelength.

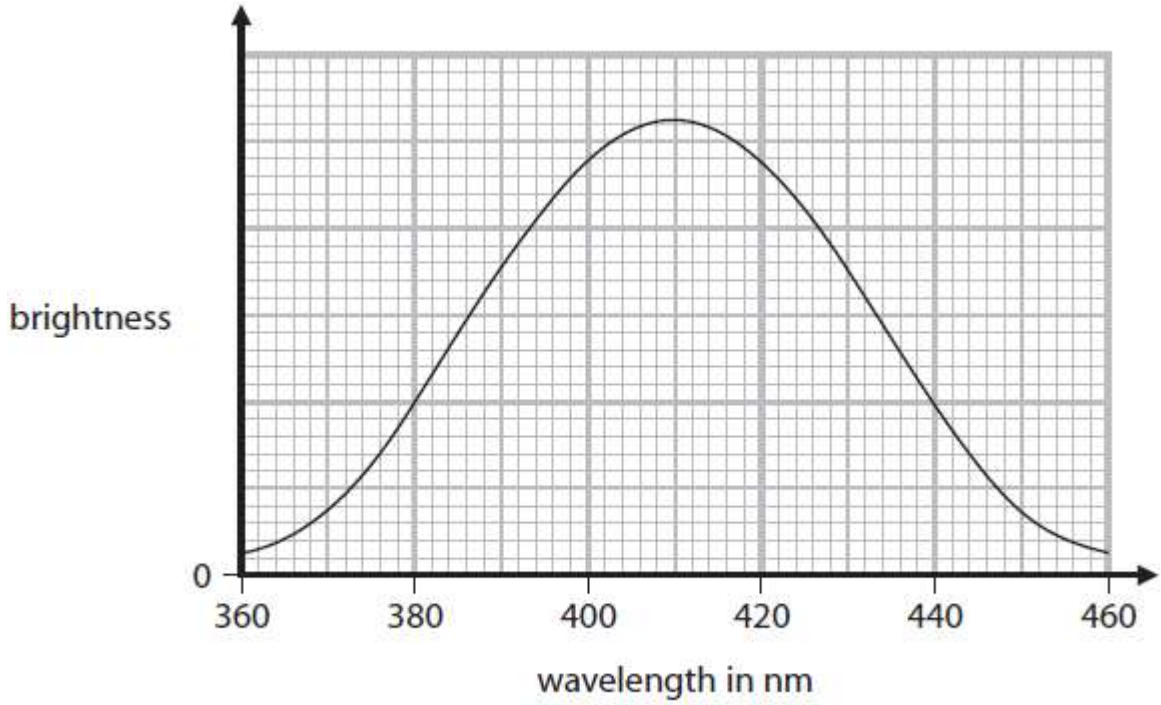

Figure 1

Describe how the brightness changes with wavelength.

(2)

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(Total for question = 2 marks)

Q5.

A hot surface emits radiation of different wavelengths.

The graph in Figure 11 shows how the intensity of the radiation emitted changes with the wavelength.

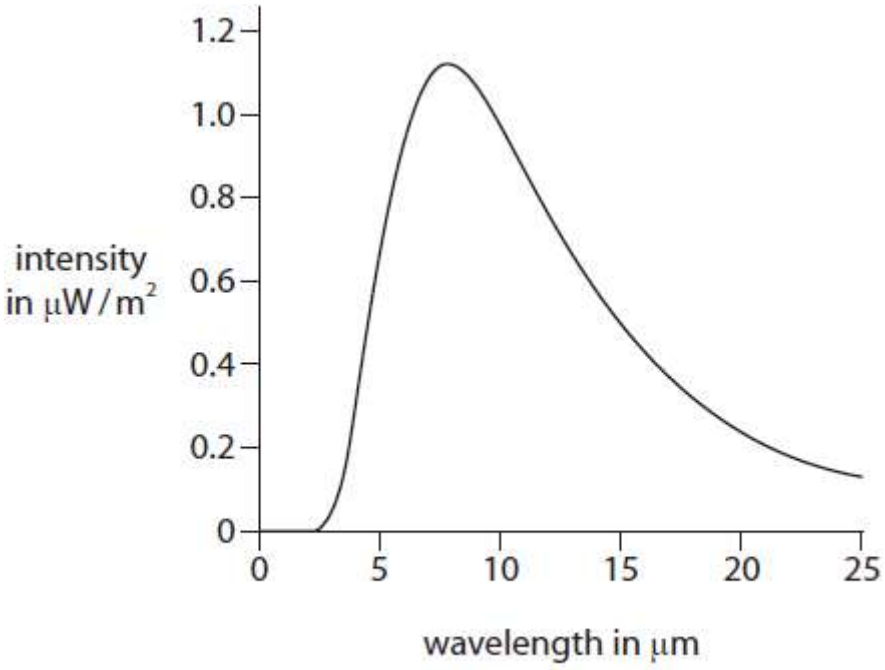

Figure 11

Describe how intensity changes with wavelength in Figure 11.

(2)

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(Total for question = 2 marks)

Q6.

All objects emit electromagnetic radiation.

The intensity and wavelength of the emitted radiation vary with the temperature of the object.

Figure 8 shows this variation for an object at two different temperatures.

The visible region of the electromagnetic spectrum is also shown.

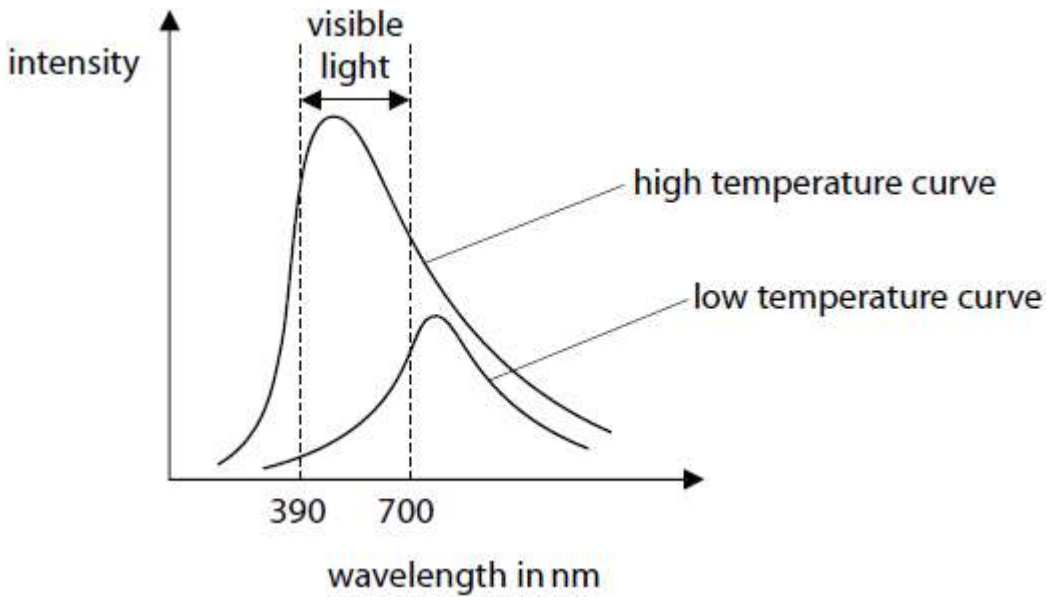

Figure 8

Describe how intensity of the emitted radiation changes with temperature.

(2)

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(Total for question = 2 marks)

Q7.

Figure 12 shows a semicircular glass block.

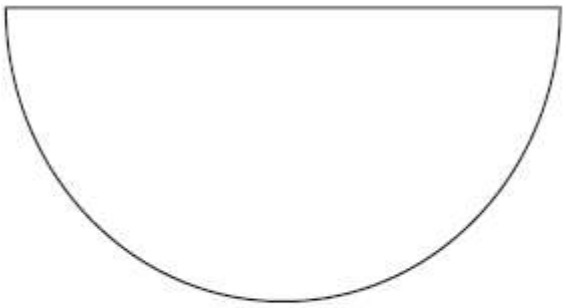

Figure 12

Describe how a student could use the semicircular glass block and other apparatus to determine the critical angle for a glass-air boundary.

You should add to the diagram in Figure 12 to help with your answer.

(4)

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(Total for question = 4 marks)

Q8.

Figure 12 shows two cans, a radiant heater and some other apparatus.

The cans absorb thermal radiation from the heater.

One can has a matte black surface and the other can has a shiny silver surface.

Both cans contain water at the same temperature.

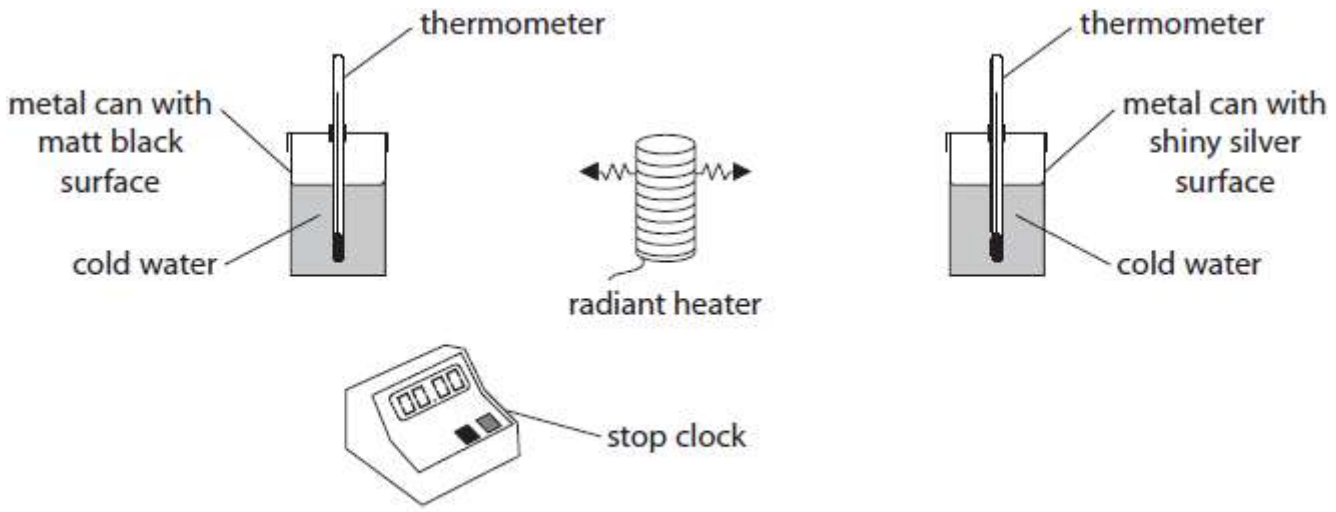

Figure 12

Describe how a student could use the apparatus in Figure 12 to determine which can is the better absorber of thermal radiation.

(4)

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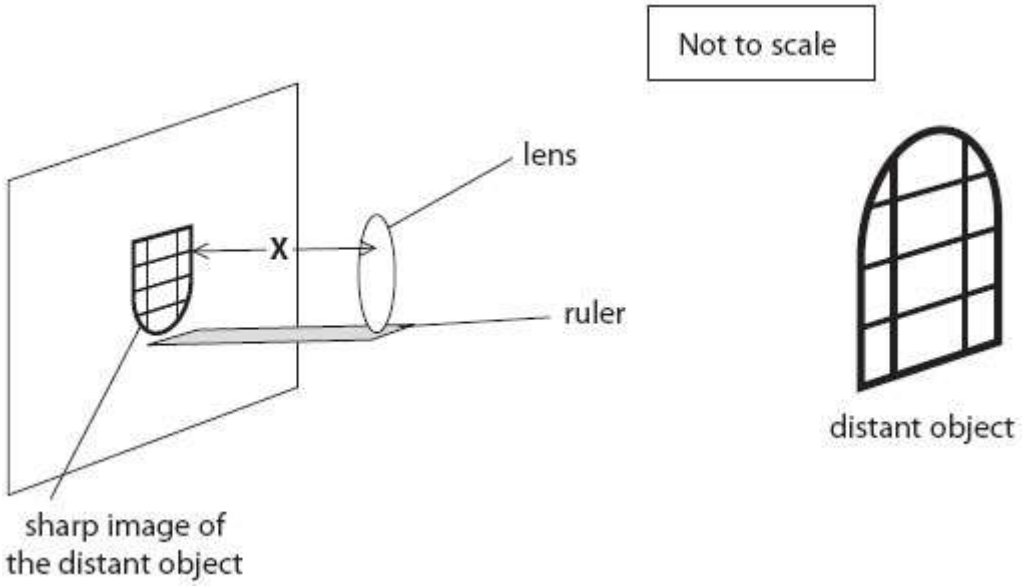
(Total for question = 4 marks)

Q9.

Some students investigate a converging lens.
The students set up the apparatus as shown.



Complete the sentence by putting a cross (☒) in the box next to your answer.

(i) The distance **X** is

(1)

☐ **A** the focal length

☐ **B** the object distance

☐ **C** the eyepiece distance

☐ **D** the magnification

(ii) Use words from the box to complete the sentences.

(2)

bigger than	diverging	real
smaller than	the same as	virtual

The size of the image is .....the distant object.
The type of image formed on the screen is a .....image.

Q10.

A student uses a converging lens to produce an image of a tree on a screen.

The image of the tree on the screen is shown in Figure 5.

The student is asked to describe the image of the tree.

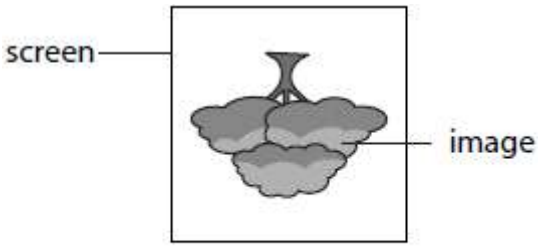


Figure 5

The student's descriptions are shown in the table.

student's descriptions of the image		correction
upside down	✓	
bigger than the actual tree	×	.....
virtual	×	.....

The first description is correct, the image is upside down.

Complete the table by adding the correct descriptions of the image.

(2)

(Total for question = 2 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Marks
	An answer that combines the following to provide a logical description of the method	Responses may be seen on diagram. allow 'calculate' for 'measure' in context ('measure angles' insufficient – must be specified)	(4)
	• measure angle of incidence (1)	measure angle from incident ray	
	• measure angle of refraction (1)	measure angle from refracted ray accept (reference to drawing path of) refracted ray (inside block) / emergent ray / ray under or leaving block	
	• measure angle(s) to the **normal** (1)	allow to 'dotted line' for normal	
	• repeat for different angle(s) (of incidence) (1)		

Q2.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO targeting: 6 marks AO1 strand 2 (6 marks)** • uses a thermometer • inserted into the can • uses a stopwatch / timer • uses a measuring cylinder / beaker with markings • sets up apparatus with lamp a fixed distance from each can • that distance is the same for each can • the measured volume is the same for each can • a starting temperature is measured • (aiming) to get the same start temperature • each, in turn, is heated for the same time • as measured on a stopwatch • a final (maximum) temperature is read • OR takes a series of temperature readings with time • and plots a graph of temperature(s) against time • the one heating up the most is the best absorber N.B. fully credit any of these in a diagram e.g. bullet points 1-8 may all be seen in a diagram	**(6)**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance

Level	Mark	Additional Guidance	General additional guidance – the decision within levels Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated statement about apparatus or procedure or the science involved	Possible candidate responses (diag?) uses/puts a thermometer in the can/water OR measures a temperature OR uses heater to heat the cans Or puts water in the cans
Level 2	3–4	Additional guidance some procedure detail but with omissions	Possible candidate responses (diag?) uses a thermometer to measure the temperature rise in the cans OR uses the heater to heat the cans for a fixed time Or uses the same amount of water in each can
Level 3	5–6	Additional guidance more detailed and structured procedure	Possible candidate responses (diag?) uses a thermometer to measure the temperature rise in the cans AND uses the heater to heat the cans for a fixed time Or uses the same amount of water in each can

Q3.

Question Number	Answer	Additional guidance	Mark
	a description including any two from window/light (from outside), lens and paper in correct order (1) move lens/paper (1) to focus (image) / to get a clear image (1)	accept lens in front/behind window and paper in front/behind lens	(2) AO2

Q4.

Question number	Answer	Additional guidance	Mark
	a description to include two of the following: increases (at first) (1) reaches a peak (1) (then) decreases (1)	is brightest at **410** (nm)	(2)

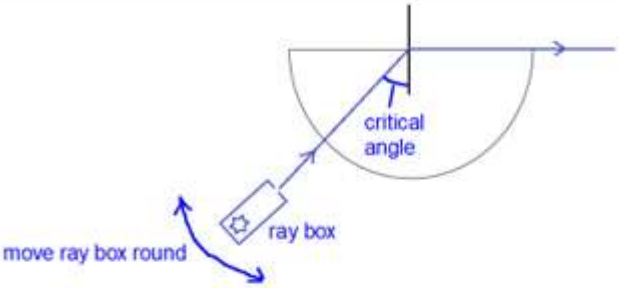
Q5.

Question Number	Answer	Additional guidance	Mark
	a description including any two from (at low wavelength) intensity increases (as wavelength increases) (1) after peak/at long wavelengths intensity decreases (as the wavelength increase)(1) intensity peaks between 7 and 10(μm) or between 1.0 and 1.2 ($\mu W/m^2$) (1)	 allow use of data from the graph to indicate the peak	(2) AO3

Q6.

Question number	Answer	Mark
	An answer that combines points of interpretation/evaluation to provide a logical description: • as temperature increases, intensity increases (1) • as temperature increases, maximum intensity occurs at a shorter wavelength (1)	(2)

Q7.

Question Number	Answer	Additional guidance	Mark
	a description to include any **four** from:		**(4)**
	shine a ray (of light) into the block (1)		
	into the block through the curved surface along a radius (1)	credit marking points in the diagram if they are clear	
	{change angle / move ray(box)} until the angle of refraction is 90° / TIR just occurs (1)		
	measure angle of incidence {when refracted angle is 90° / when TIR just occurs} (1)	allow 'calculate' for 'measure' plot angle i against angle r	
	repeat measurement of critical angle (1)	if light only enters block at straight edge, maximum 1 mark (for MP1)	

Q8.

Question Number	Answer	Additional guidance	Mark
	a description including any four from **one** method **Using temperature change method:** any one control (1) start the stopclock and switch on the heater (at the same time) (1) measure over a certain time (1) measure final temperature of each (1) largest temperature difference/rise gives better absorbing surface (1)	allow testing each can separately (two experiments) same distance from heater same volume/mass of water allow same volume/surface area of cans lids on both cans start the stopclock and measure initial temperature (at the same time) Accept 'highest temperature is better'	**(4)** **AO1**

	Using time change method: any one control (1) switch on the heater and start the stopclock (at the same time) (1) measure time taken (1) to reach a certain temperature (1) quickest time to reach a certain temperature gives better absorbing surface (1)	same distance from heater same volume/mass of water allow same volume/surface area of cans lids on both cans accept 'fastest (to reach temperature) is better'	

Q9.

	Answer	Acceptable answers	Mark
(i)	A the focal length (1)		**(1)**
(ii)	smaller than (1) real (1)		**(2)**

Q10.

Question number	Answer	Additional guidance	Mark
	diminished/smaller (1) real (1)	accept recognisable spellings e.g. reel do not credit not virtual	**2** **AO2.2**